

A Focused Animation Curriculum Model

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Figure 1: Student projects: *Off Track*, intro short film ©Catherine Hackl, Ella Peterson, Ivery Chen, Sarah Jihyun Woo, and Zackary Entwistle; *Dim Sum*, intermediate project ©Kelly Wang, and *Raindiance*, independent project ©Allison Yeh

ABSTRACT

We offer this curriculum as a model for a focused animation program that does not require the infrastructure of an animation major, multiple instructors, or many courses. Students who have taken two courses are well-prepared for technical director positions. Many students report this experience as a college highlight.

The success of the program is due to several key practices. Group critiques mimic “dailies” in industry and allow students to improve work in progress. Students are motivated to learn animation basics when they are applied to a final short film. We emphasize problem-solving over recipe-following so students become self-sufficient in learning new techniques. We provide a supportive and enthusiastic structure through in-class activities, TA support, and a well-attended final showcase. We discuss these principles along with course aims and lessons learned as we refined our approach.

CCS CONCEPTS

• **Social and professional topics** → **Computer science education**; • **Human-centered computing**; • **Applied computing** → **Arts and humanities**;

KEYWORDS

animation, teaching, production, curriculum

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1 INTRODUCTION

Many undergraduate universities (such as Univ. of Washington, USC, and U. Penn) and art schools (such as Ringling, CalArts, and Emily Carr) offer rich computer animation curricula such as those described in [Ebert and Bailey 2000] and [Comminos et al. 2009] that are ideal for in-depth study. Alternatively, we offer a focused and intense animation experience. Our program, taught by one instructor for nearly two decades, has two core courses that follow the animation industry production pipeline based in part on our professional visual effects experience. An occasionally-taught third course offers the opportunity to work on one larger production as a group, and students may also do independent work.

2 COURSES AND LEARNING OBJECTIVES

The two core courses share the following aims.

Learn and use the production pipeline to create an animated film. Students do projects in modeling, shading, lighting, and animation to learn enough skills to make a short film in the first term. The second course adds character rigging and advanced topics such as simulation and compositing.

Develop the ability to critique and improve work in progress. To recreate the intermediate feedback that we valued in our production career, each project has in-progress and final due dates during which work is critiqued in class by students and the instructor. Students learn techniques from each other, develop their critical eyes, and build trust in the peer review process.

See animation as an artistic and personal expression. We view films created with a variety of animation techniques. We discuss the expressiveness of pieces even if the primary objective is technical. Animation is a painstaking art form; we impress upon students from the earliest stages they should choose subjects that express their feelings so they have a personal stake in the outcome.

Become self-sufficient in learning computer animation techniques. We provide tutorials and guidance when students are learning

the pipeline and software. In the second course, we move toward independent learning. This gradual approach offers security when needed and freedom as students explore more disparate topics.

2.1 Curriculum

We introduce Autodesk Maya software in **Introduction to Computer Animation** through short projects in each pipeline area. In parallel, students work on pre-production tasks for a group short film project. When these two tracks are completed, they move into full production of the shorts.

After their film success, students are hooked, but wish they could redo some parts with better outcomes. **Intermediate Animation** emphasizes proper workflows and technically correct implementations. For the final project, students explore one topic of their choice in depth and create portfolio-quality demos.

In **Advanced Animation**, students make a polished group film. After several trials, we determined structural requirements for the successful completion of the film which are described in Section 4.

Students who have completed one course may create one or more **independent study** courses. These are satisfying to advise because students are motivated to pursue their choice of topic and they assume the burden of project structure.

3 KEY PRACTICES AND PRINCIPLES

We believe the success of our curriculum lies in several key practices honed over time. **Critique of work in progress** offers students practice in evaluating their own and other students' work and determining and implementing improvements before final evaluation. We teach Feldman's four-stage critique model [Feldman 1994]. Students practice using the model on past student work so that even those without prior critique experience have the same foundation. Work is critiqued in class at both the progress and final stages so students can observe and evaluate modifications.

Maya is well known to have a steep learning curve so we adopt an "eat dessert first" philosophy. We focus on **early gratification** by teaching the skills needed to produce the final short film without theory, math, and other "under the hood" topics that might bog students down. If we insisted on correct technique, they would never finish their films and perhaps not continue with animation. We leverage previous skills in storytelling and visual design and increase their stake by asking for a personally expressive piece. We do steer the projects to ensure they are doable in the time given.

In early years, we used book and video tutorials to teach the software, but found these resources ignored problem-solving. We supplemented with lectures that covered the theoretical basis of methods, but students did not connect theory and practice. Over time, we learned to **prioritize problem-solving over recipes or button-pushing** by authoring our own tutorials that motivate choices by stating the desired outcomes. We present and weigh options at decision points and model the curious problem-solving behavior we hope students will adopt. As they progress through the curriculum, they do more of their own research and experiments and gradually become comfortable learning independently.

Our last principle is to **create a supportive and enthusiastic environment**. We engage in fun collaborative story and art activities. Students receive extensive written feedback on their work

even when it is excellent. Late penalties are not severe to motivate finishing projects. We produce a well-attended final showcase of student work followed by a wrap party that feels like a premier. Throughout all the courses, we discuss mental health challenges such as long hours, flaky software, and personal feelings of inadequacy to normalize them and build community. In early years, we used a more punitive "sink or swim" approach, but realized the students could not draw on past experience when starting out so we moved to a learning on-the-job model. As a result, the quality of final projects increased substantially, students felt supported, and we enjoyed the increased engagement.

4 LESSONS LEARNED

In addition to the evolution of principles described in the last section, we offer additional lessons learned.

The advanced course emerged after a few unsuccessful attempts. In one, students formed a Group Independent Study Project but without strong leadership or a production blueprint, they did not finish a film. In another effort, four students did simultaneous independent studies focused on one project. They created many assets, but fell short because their scope was too big.

The next iteration, a formal course, featured a solid narrative and more students, but lacked leadership so decisions were endlessly debated. From this experience, we devised these requirements:

- A minimum number of students who can work in Maya
- Roles: director, producer, editor, composer, sound designer
- Defined leadership and working roles for pipeline areas
- Preproduction tasks completed before the course begins

Personally, we learned that the instructor can be a director or producer, but not both. In 2016, we successfully created *Toymaker* [Polhemus 2017], which had a successful run at many festivals.

Finally, a cost of our approach is being selective in what we teach, only briefly covering character animation, simulation, visual effects, and software other than Maya, though students can pursue these topics independently. Our curriculum is also applicable to virtual reality, games, and other real-time media.

5 CONCLUSIONS

We measure our success by the consistently excellent work of students. Diligent students who pursue production careers are usually successful landing jobs at studios such as Pixar, Disney, ILM, Electronic Arts, and Laika. In course surveys, they often report these courses to be the most rewarding of their college careers.

We are grateful for our department's support of our arts-related courses. Animation dovetails well with our other graphics, vision, VR, and game engine offerings and students who pursue both paths report high satisfaction with the art-science mix.

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